Lab 1 Report

System Identification

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1 Introduction

The aim of this lab is to determine the specific system model of ... using ... methods. This is called System Identification...

2 System Modelling

What happens in this section? ...

2.1 Methodology

1. First, the list was created.
2. Subsequently, steps were added
3. Lastly, I got tired of writing this method.

2.2 System Model

2.2.1 A differential equation describing the motion of the mass

Explain it

$$50m3 + terrible = maths - \frac{dy}{yay} \tag{1}$$

2.2.2 A transfer function derived from the differential equation

How'd you do it? What does it represent? From Equation 1 we get...

$$transfer_function here \tag{2}$$

3 System Identification

3.1 Step Response Test

Damping Coefficient	**Spring coefficient**
12345	12345

Table 1: System Identification Values from my brain

The transfer function below is obtained...

$$transfer_function here \tag{3}$$

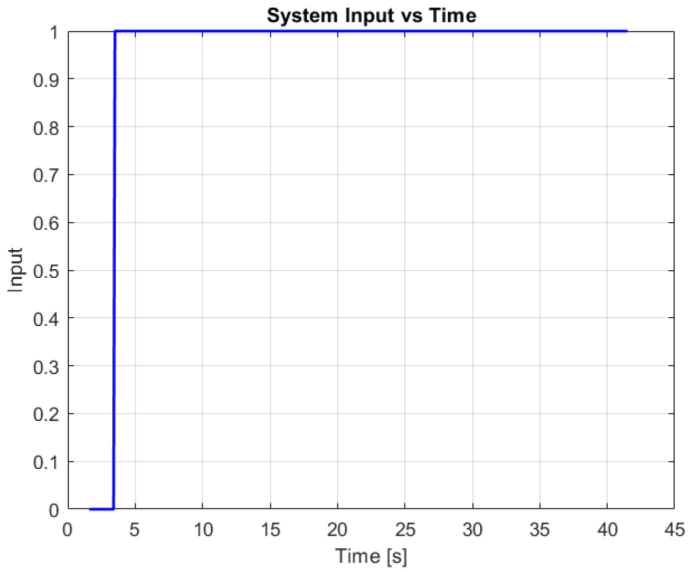

Figure 1: Step input

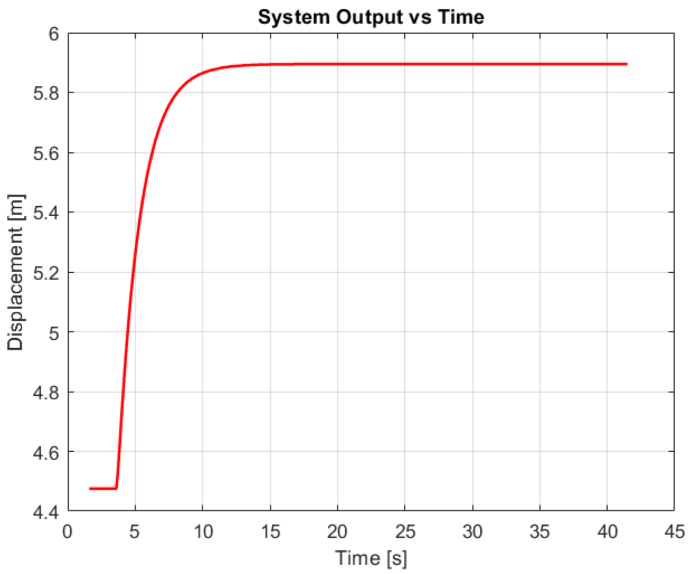

Figure 2: Step response

Calculation of the System Identification Values: How did you get the above values? Elaborate more than you would in methodology. Remember to include formulae, etc.

3.2 Frequency Response

Why these frequencies? Expand on the previous section.

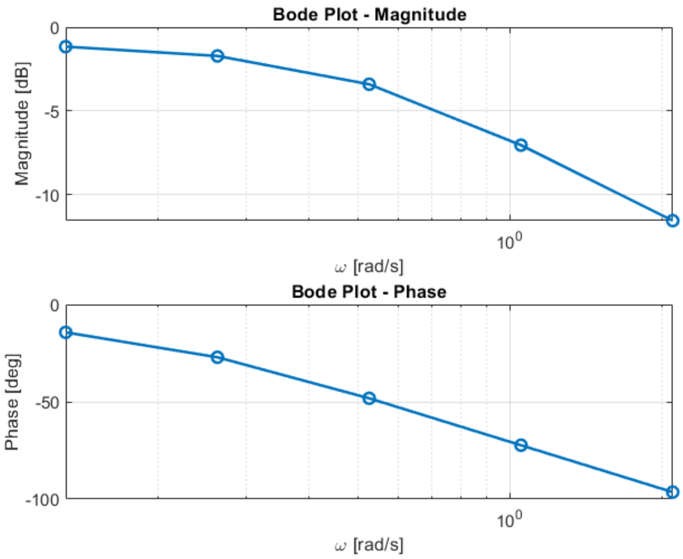

Figure 3: Frequency response

3.3 Model Validation

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How? Why? What were your findings?

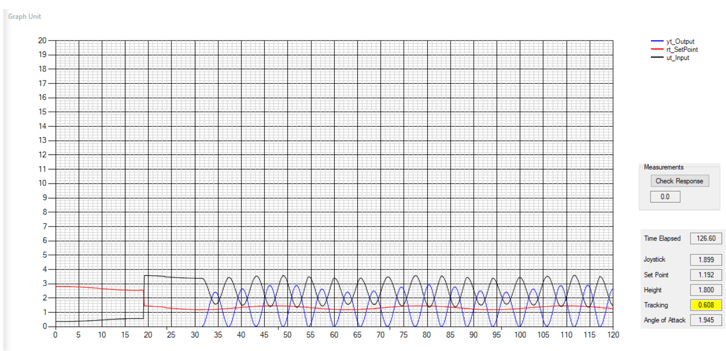

Figure 4: This is a random image, use your own to show some sort of validation or comparison

4 Conclusion

Rename things/sections as you see fit. Please still refer to the lab manual about what else to include and the rubric for what you'll actually get marks for. Be concise.

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A Appendix and Notes

Here you can add code or other relevant things that weren't very applicable within the report. Put things in here that you actually reference in the report.

A.1 MATLAB Code

```
% Example code snippet
t = 0:0.01:10;
y = sin(t);
plot(t,y);
```

A.2 Additional Figures

Add extra supporting figures here if needed.